Journal Data Analysis

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Module: MATH6183-Data Mining and Analytics

Dec 2025

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1 Introduction

Initial inspection of the dataset suggests that research topics may change over time, and citation counts are likely not well explained by linear relationships with observable paper characteristics. This report analyses a collection of papers drawn from multiple journals spanning applied mathematics and operational research. The objective is to formulate hypotheses related to citations and journals and evaluate them by examining both textual and non-textual characteristics of the data, identifying latent thematic structures through topic modelling, and applying predictive models to assess citation behaviour and journal classification.

2 Initial cleaning & non-text data exploration

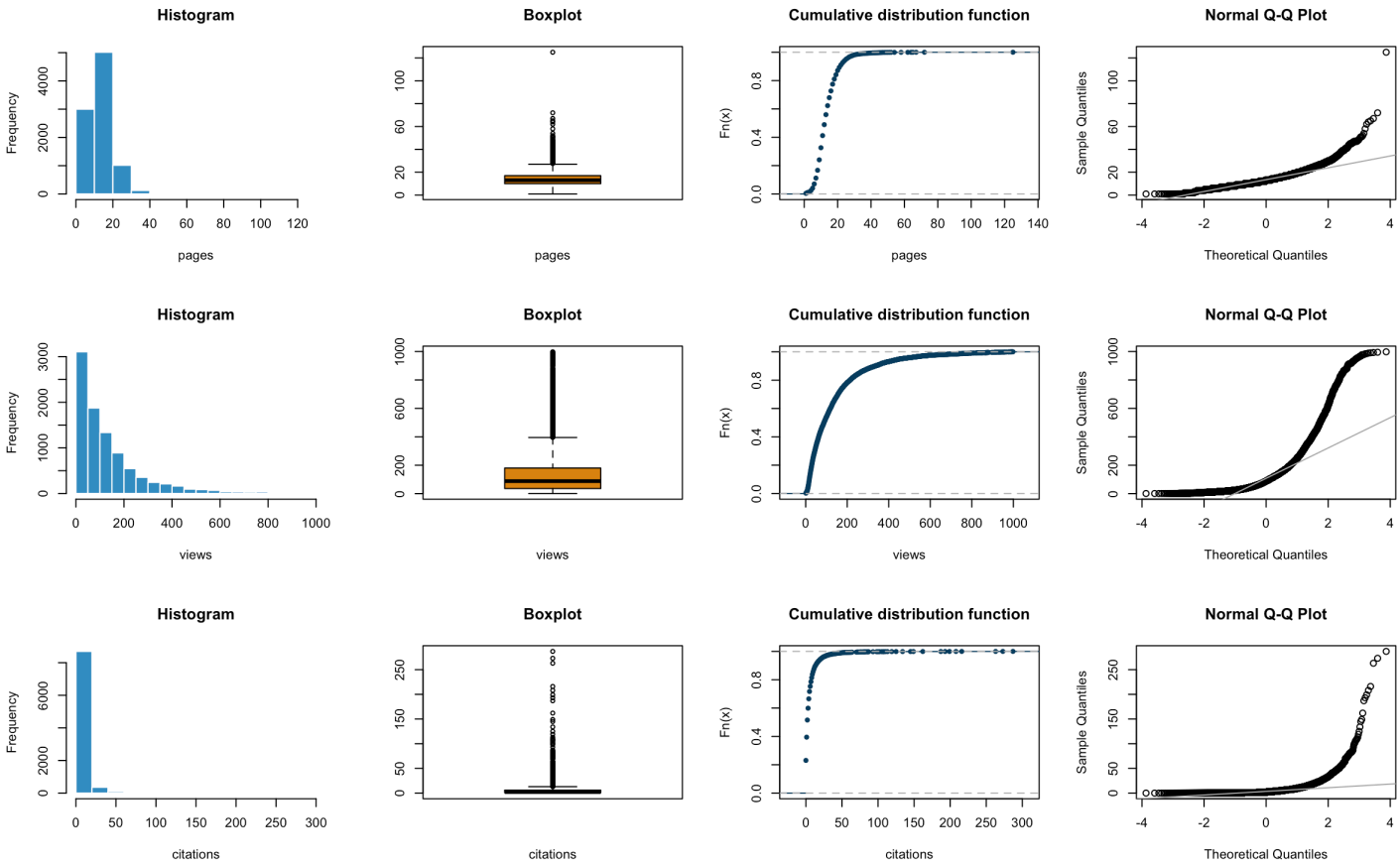
2.1 Univariate analysis

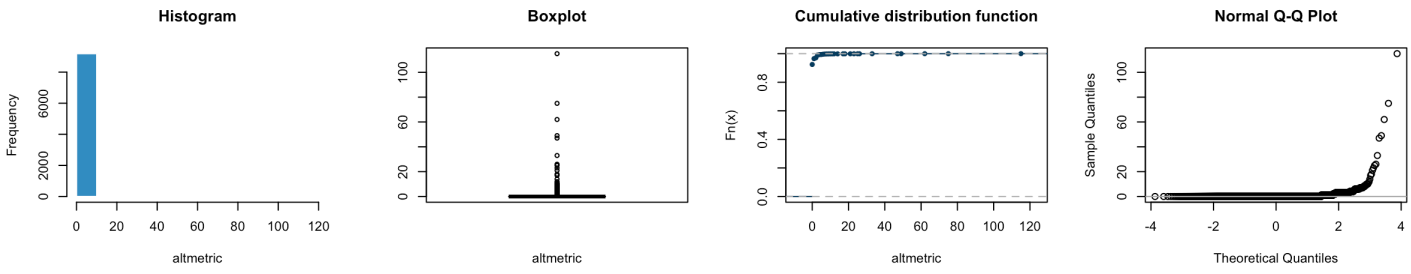
Univariate analysis is conducted to summarise the distributions of individual variables and to identify basic structural patterns in the dataset. This step provides an initial understanding of data characteristics, such as skewness, outliers, and imbalance, which informs subsequent preprocessing choices and model selection.

2.1.1 Numerical variables

In this dataset, the numerical variables comprise page count, views, citations, and altmetric score. The views variable contains 183 missing observations (approximately 2% of the dataset). Since many machine learning algorithms could require complete data, the proportion of missing values is small, and imputing values may introduce bias, these missing values are excluded from the analysis.

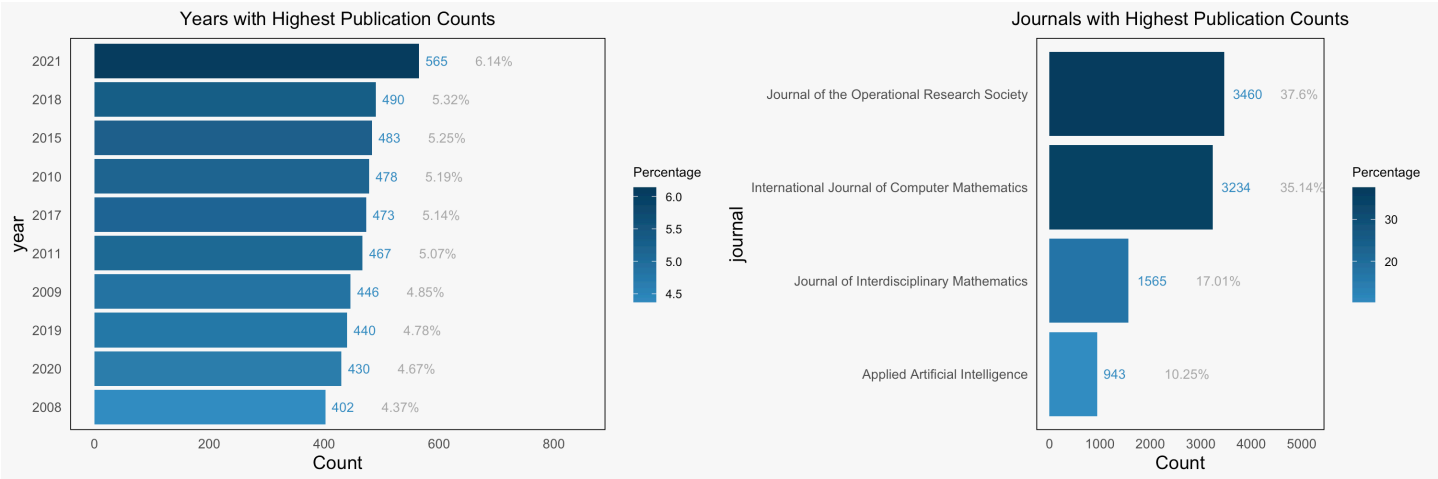
Initial examination of numerical variables reveals strong right skewness and outliers, which can affect model performance. Because of this, logarithmic transformations are applied to stabilise variance and reduce the influence of extreme values.





2.1.2 Categorical variables

The data are relatively evenly distributed across publication years, with a peak in 2021. In contrast, the journal distribution is highly imbalanced, with approximately 73% of articles belonging to the Journal of the Operational Research Society and the International Journal of Computer Mathematics. This class imbalance suggests that corrective strategies, such as resampling or class-weighted models, may be necessary to mitigate potential bias in the classification task.

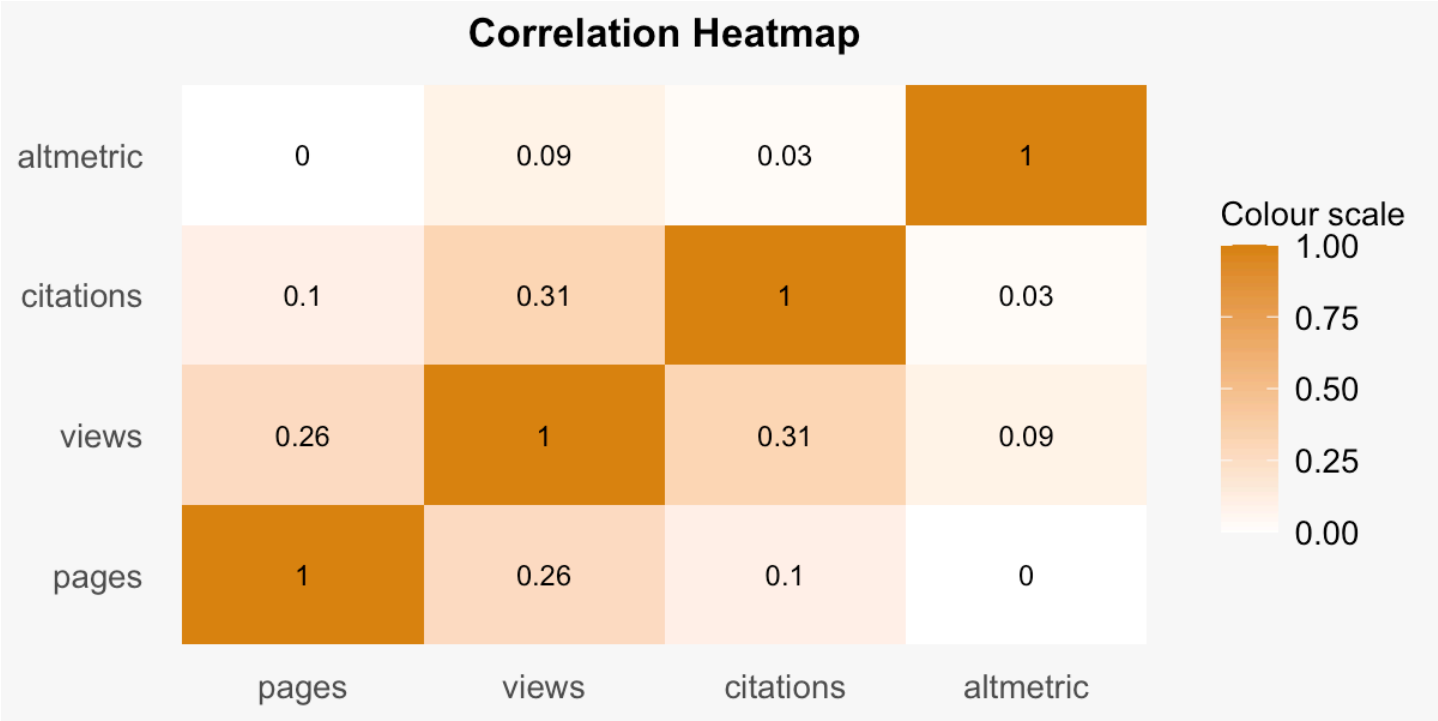



```
======== Variable:  year ========
Unique value count:  23
Mode:  2021
======== Variable:  journal ========
Unique value count:  4
Mode:  Journal of the Operational Research Society
```

2.2 Bivariate analysis

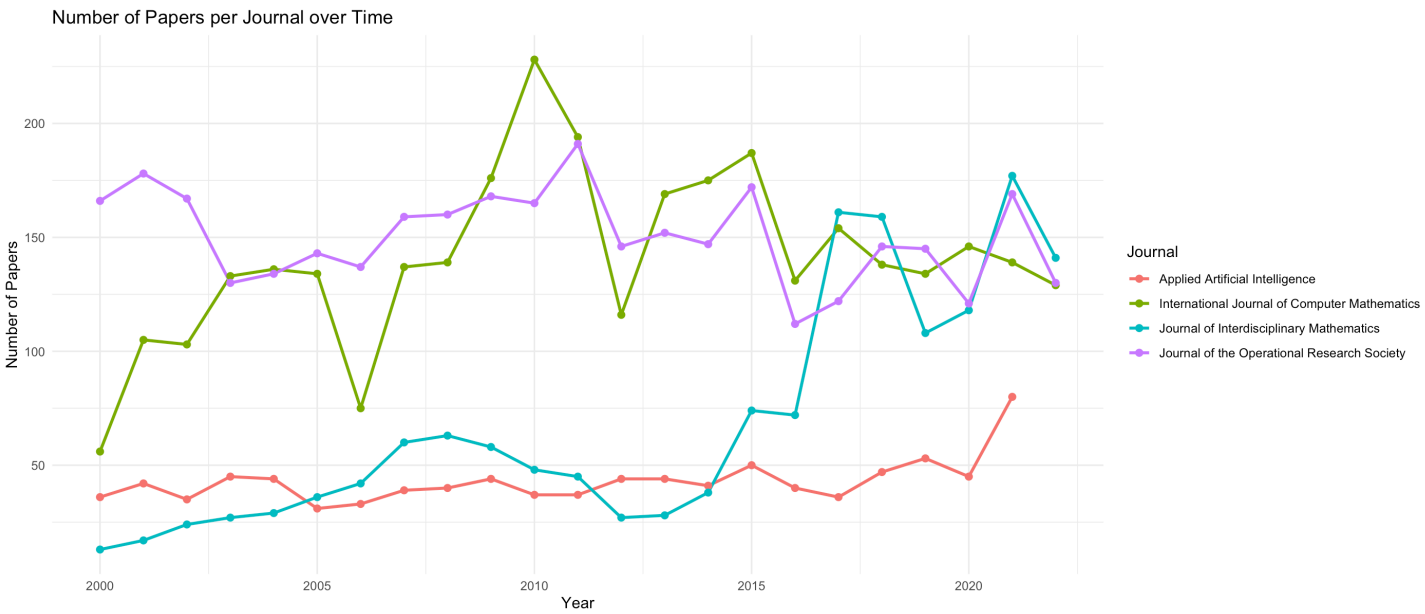
2.2.1 Numeric × Numeric

Overall, no strong linear relationships are observed among the numerical variables. The highest correlation is found between citations and views (0.31), indicating a relatively weak positive association. Initial observations suggest that linear dependencies are limited; therefore, potential relationships among independent and dependent variables are almost certainly non-linear. Consequently, models capable of capturing non-linear interactions are probably considered in subsequent analyses.



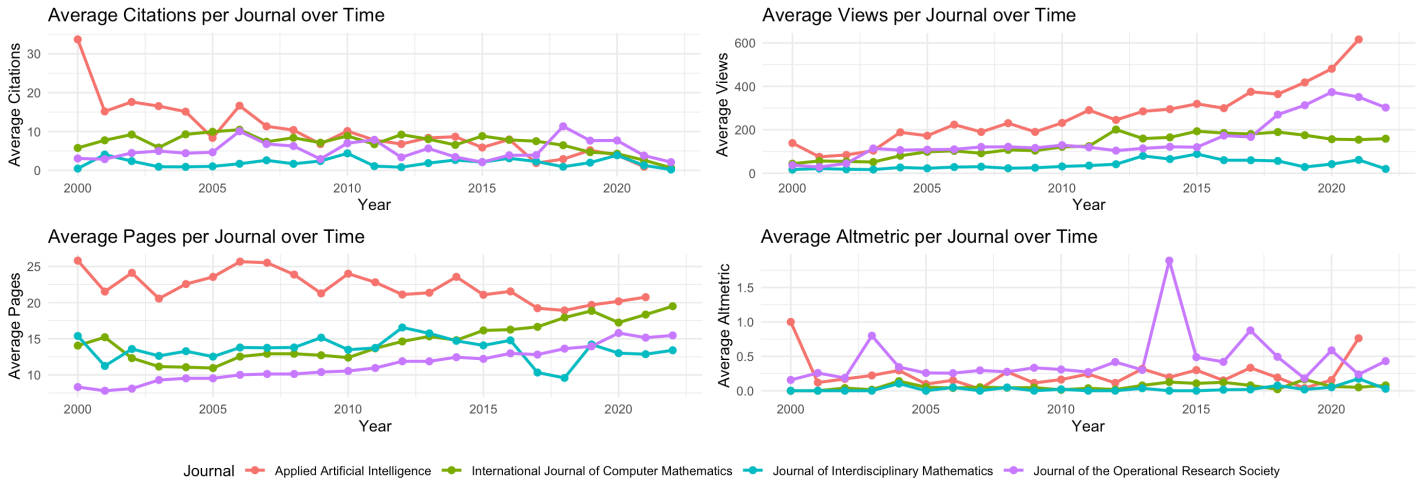
2.2.2 Categorical × Categorical

The graph shows a consistently high publication volume for the Journal of the Operational Research Society, averaging around 150 papers per year and ranking first overall. What is interesting in this chart is the growth of the Journal of Interdisciplinary Mathematics, particularly after 2015, surpassing other journals in publication volume in several recent years. These patterns have implications for both classification and regression models, as they may affect model generalisation and the stability of learned relationships across different periods.



2.2.3 Numeric × Categorical

Average citations for Applied Artificial Intelligence show a declining trend over time, whereas average views increase steadily, suggesting a growing readership despite reduced citation intensity. Across journals, average page counts show a gradual upward trend, indicating a shift toward longer and potentially more detailed articles. Altmetric attention remains relatively low and volatile, implying limited public engagement. While these patterns may suggest that citations, views, and page length are likely informative features for predictive modelling, this analysis remains descriptive. Formal hypothesis testing or causal analysis would be required to assess whether and how these variables directly influence one another.



3 Hypothesis framing

While mathematics-related publications increase over time, the thematic drivers of this growth are not immediately evident. We therefore hypothesise a post-2015 shift toward numerical and computational topics (H1). Additionally, weak linear relationships between citations and explanatory variables motivate the hypothesis that citation behaviour is inherently complex (H2).

4 Text analysis

4.1 Text preprocessing

Text preprocessing is a crucial first step to ensure that the extracted features are informative rather than noisy. Standard normalisation procedures are applied, including the removal of stop words, punctuation, special characters, HTML tags, numerical tokens, and mathematical expressions. All text is converted to lowercase and lemmatised to standardise lexical forms, and additional general terms with limited semantic value are removed. These steps reduce lexical sparsity in the TF–IDF representation, likely promoting the extraction of meaningful semantic patterns rather than surface-level variations. Stemming is not applied, as it mechanically truncates words and may obscure semantic meaning, whereas lemmatisation might preserve contextually meaningful word forms.

4.2 Text representation

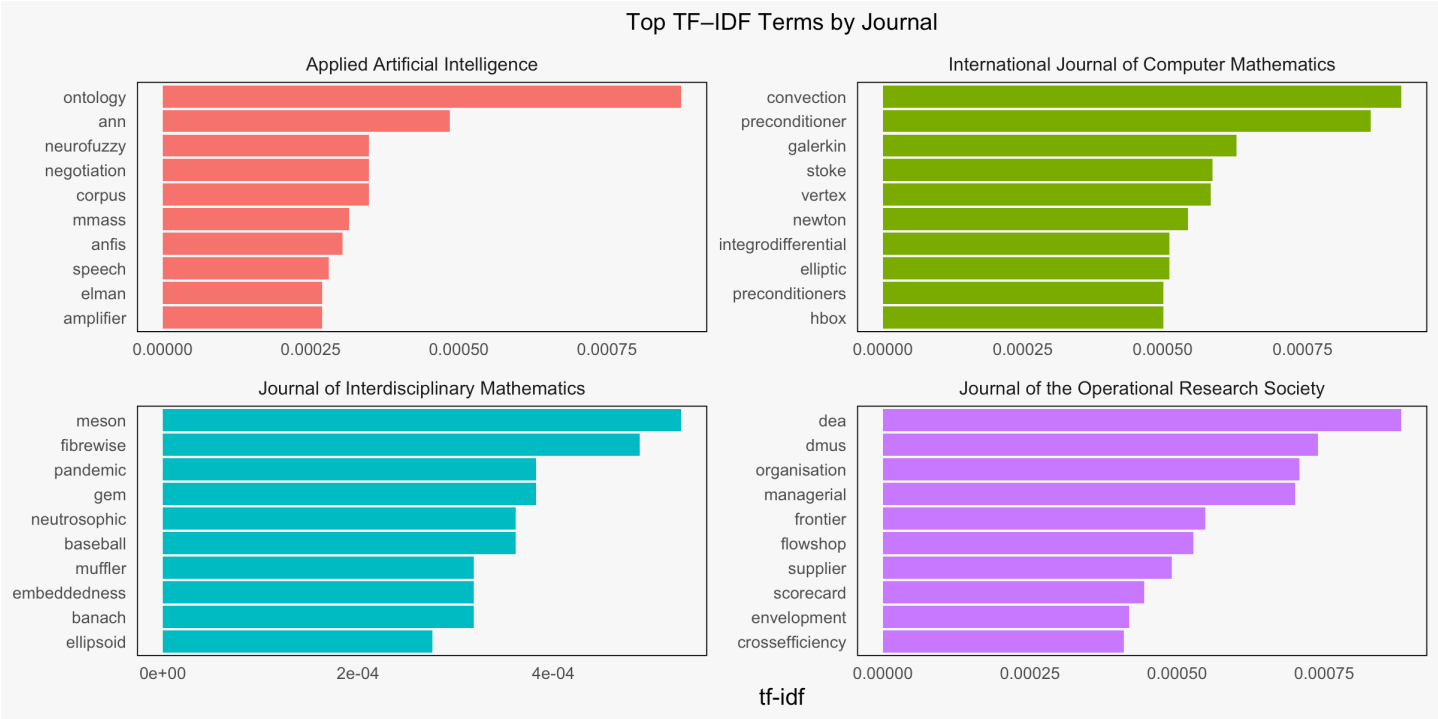
After preprocessing, the text is transformed into numerical representations suitable for modelling. Two vectorisation approaches are employed in this analysis: the bag-of-words representation is used for the LDA topic model, while TF–IDF is applied in classification, association rule mining, and clustering tasks.

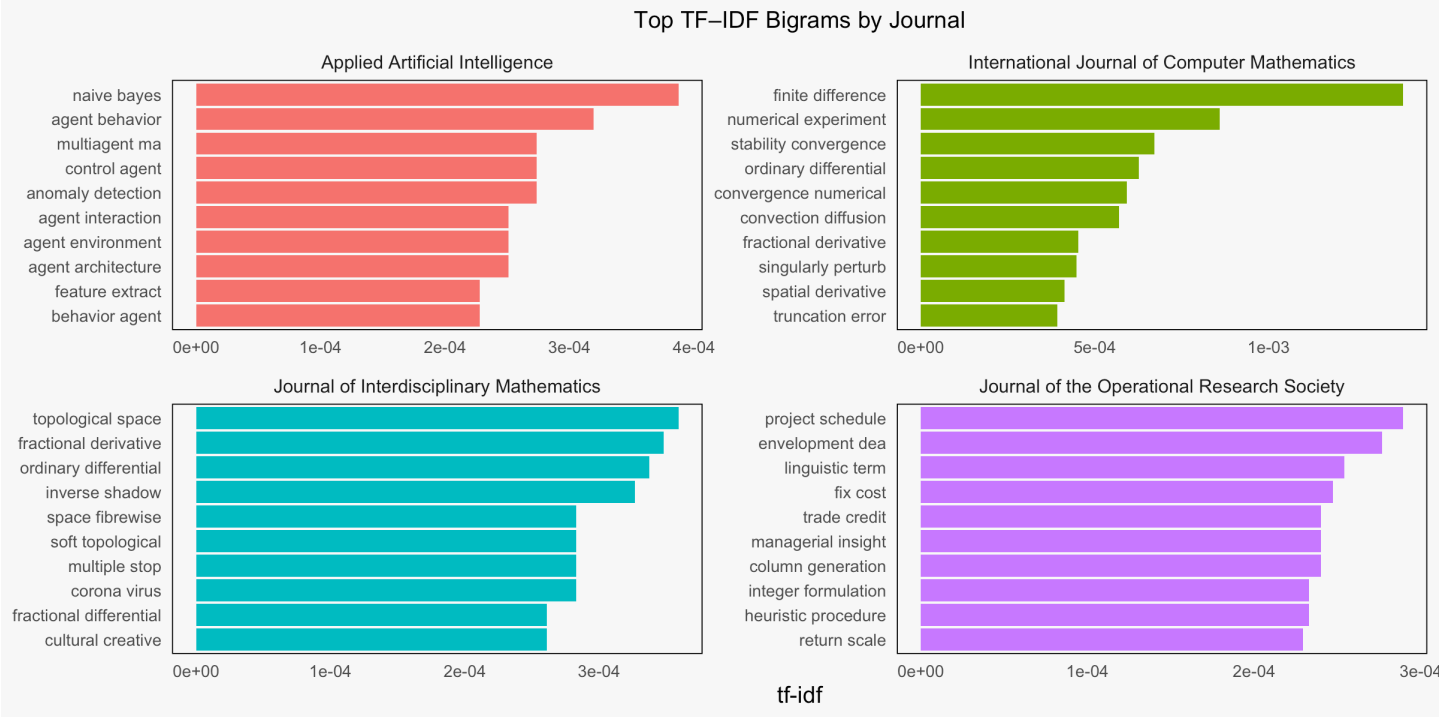
4.2.1 Bag of words

The bag-of-words model in NLP is a widely used method to create a numerical dictionary for computers to identify patterns between documents. It is widely adopted because of its simplicity and interpretation. However, the limitation of this approach is that it ignores contextual information and can lead to highly sparse representations as corpus size increases, increasing computational cost. To mitigate these limitations when fitting the LDA model, only terms appearing in at least 100 documents and in no more than 70% of documents are retained.

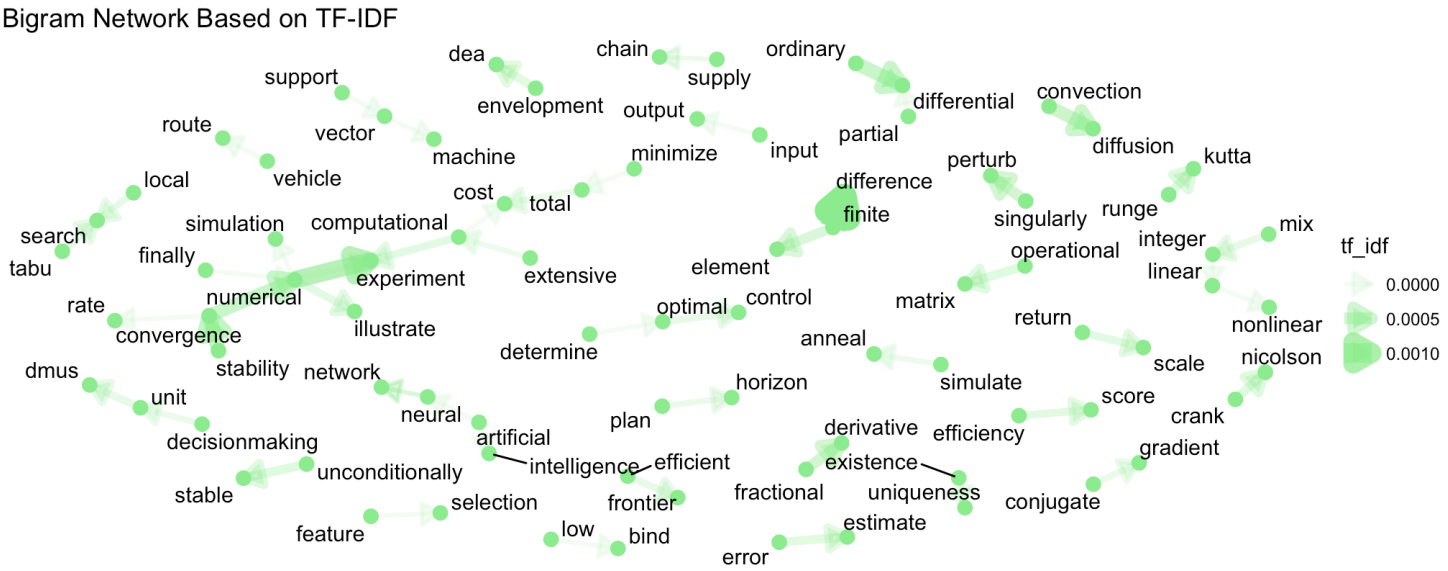
4.2.2 Term frequency & TF-IDF

TF–IDF is used to highlight journal-specific vocabulary by downweighting common terms. Unigram and bigram analyses reveal clear lexical differentiation across journals, with mathematically oriented and optimisation-focused journals illustrating more concentrated vocabularies, while the Journal of Interdisciplinary Mathematics shows a broader distribution. These patterns support H1 by indicating differing thematic scopes across the Journal of Interdisciplinary Mathematics and motivate subsequent analysis of which topics contribute to publication growth, although TF–IDF remains limited to surface-level term importance.





The TF-IDF-based bigram network shows word pairs that often appear together and are likely more important, and may reveal meaningful information. These structured lexical groupings support the validity of the LDA topics by demonstrating coherent and domain-specific vocabularies.



5 Topic Modelling and Predictive Analyses

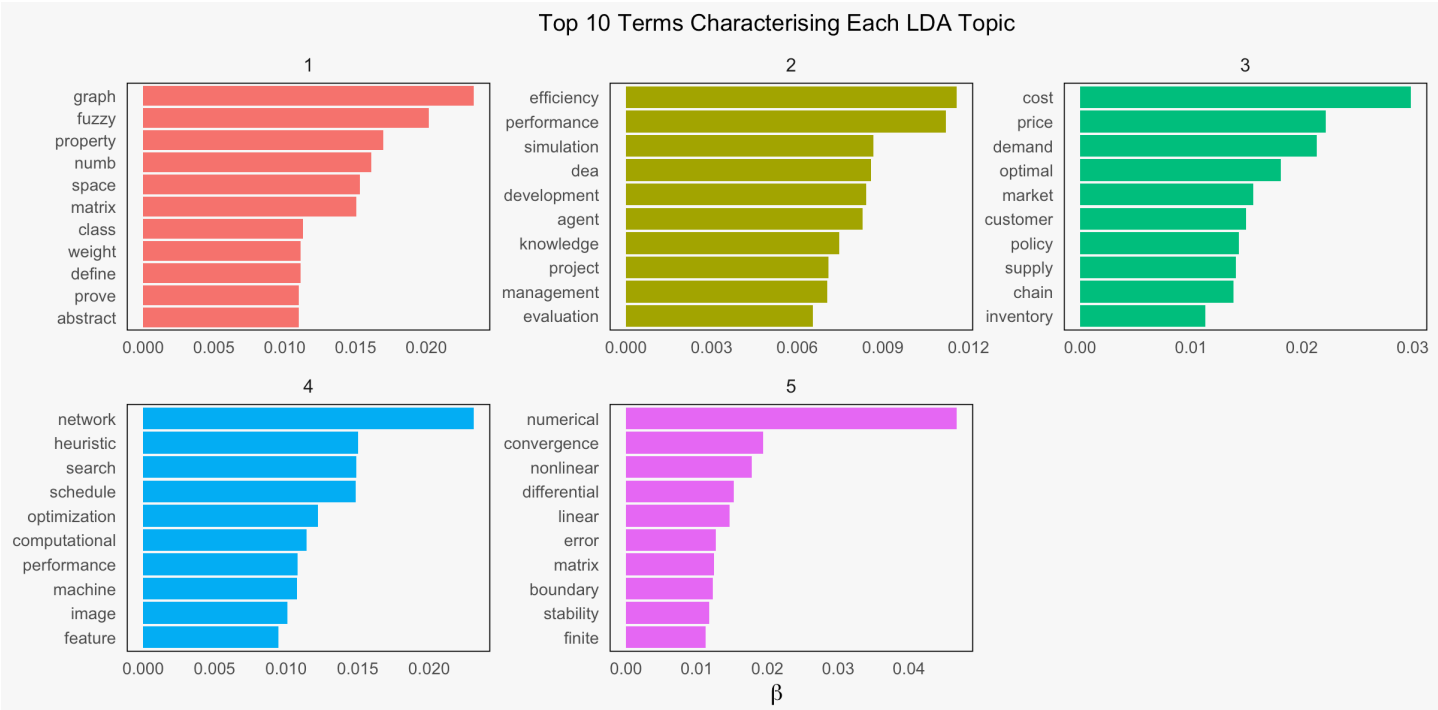
After cleaning, the dataset was reduced from 9,385 to 9,202 documents, while the vocabulary size decreased by around 13%, probably improving model stability and reducing noise.

5.1 Topic modelling

Following LDA estimation, topics were successfully assigned to 9,197 documents. Five documents were excluded after final preprocessing steps due to invalid abstracts in which nearly all tokens consisted primarily of stop words, HTML tags, and irrelevant special characters, preventing meaningful topic assignment.

Overall, the topics align well with major research streams in applied mathematics and operational research, though some lexical overlap suggests gradual rather than sharply separated thematic boundaries.

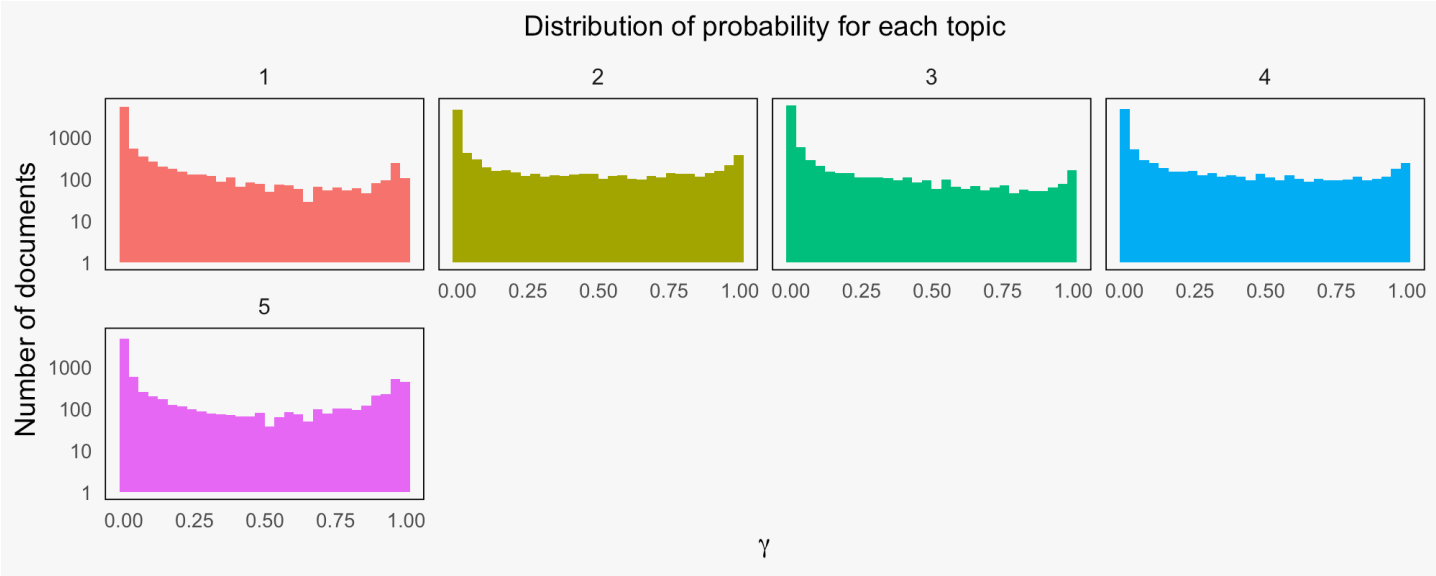
The LDA model identifies five distinct topics with coherent thematic structures. Topic 1 is associated with fuzzy systems and graph-based representations, indicating theoretical modelling and abstraction. Topic 2 emphasises performance evaluation and decision analysis, reflected by terms such as efficiency, DEA, and management. Topic 3 captures economic and supply-chain considerations, with dominant terms including cost, price, and demand. Topic 4 relates to heuristic and computational optimisation, highlighting search, scheduling, and network methods. Topic 5 focuses on numerical analysis of differential systems, characterised by convergence, stability, and finiteness.



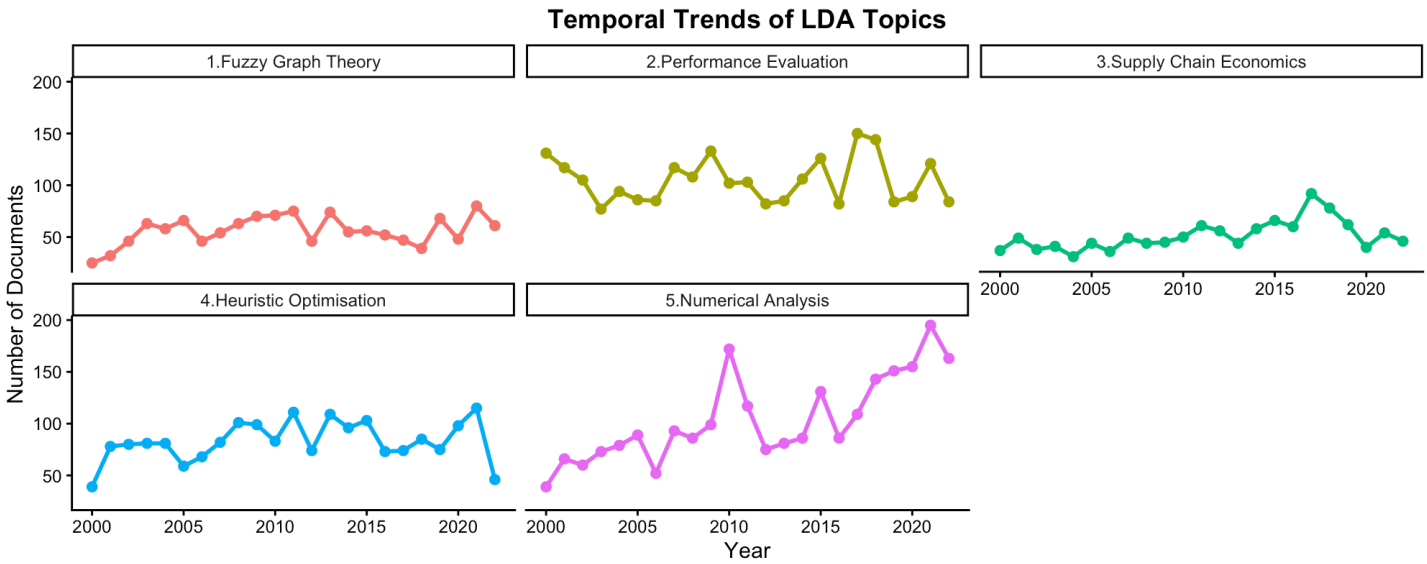
After evaluating multiple choices of K, the dataset can be categorised into five main topics: Fuzzy Graph Theory, Performance Evaluation, Supply Chain Economics, Heuristic Optimisation, and Numerical Analysis.

Topic	Topic_name	representative_words
01	1.Fuzzy Graph Theory	graph fuzzy property numb space matrix class weight define prove
02	2.Performance Evaluation	efficiency performance simulation dea development agent knowledge project management evaluation
03	3.Supply Chain Economics	cost price demand optimal market customer policy supply chain inventory
04	4.Heuristic Optimisation	network heuristic search schedule optimization computational performance machine image feature
05	5.Numerical Analysis	numerical convergence nonlinear differential linear error matrix boundary stability finite

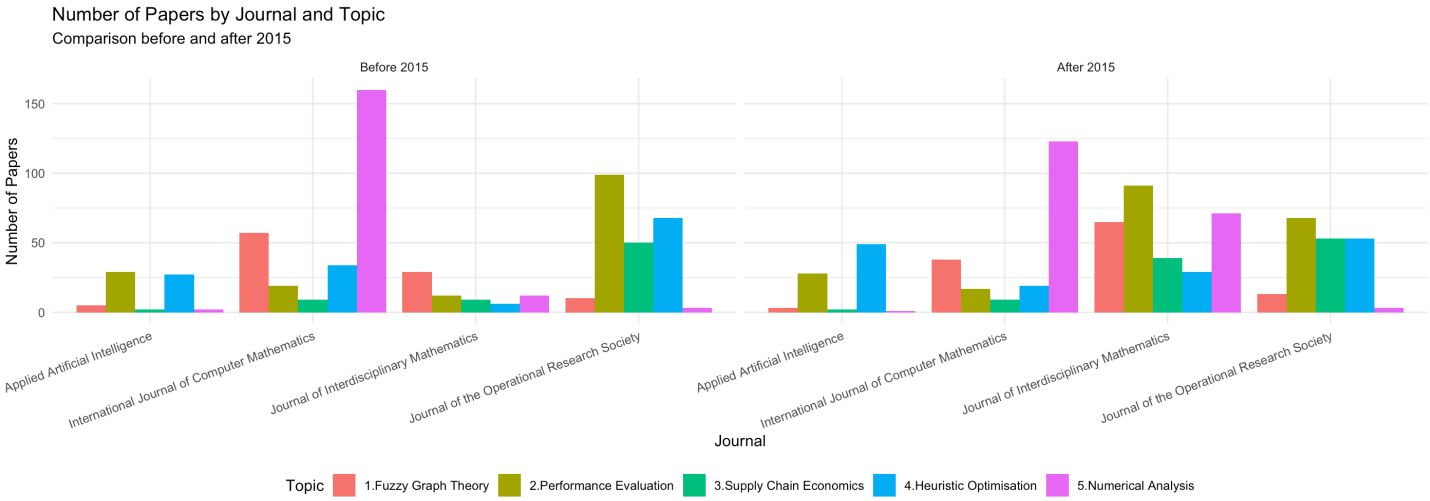
The distribution chart below shows that documents are relatively evenly distributed across topics. These results would seem to suggest that the LDA model is able to capture specific themes based on prominent word patterns. Nevertheless, many documents fail to demonstrate strong dominance by any single topic, indicating overlapping thematic structures and ambiguous topic assignments.



The temporal distribution of topics might reveal distinct developmental patterns, providing direct support for the thematic evolution hypothesis. Topics 1 to 4 display relatively stable publication trends with moderate fluctuations, suggesting established and consistently researched areas. In contrast, Topic 5 shows a pronounced upward trend, particularly after 2015, indicating increasing research attention, likely driven by growing interest in numerical and convergence-related methods.



The distribution of papers by topic and journal shows clear journal-specific patterns before and after 2015. While the International Journal of Computer Mathematics remains closely associated with numerical analysis, its dominance declines after 2015. This shift coincides with increased contributions from the Journal of Interdisciplinary Mathematics, whose focus shifts toward performance evaluation and numerical analysis, supporting the hypothesis of thematic evolution over time.



5.2 Regression

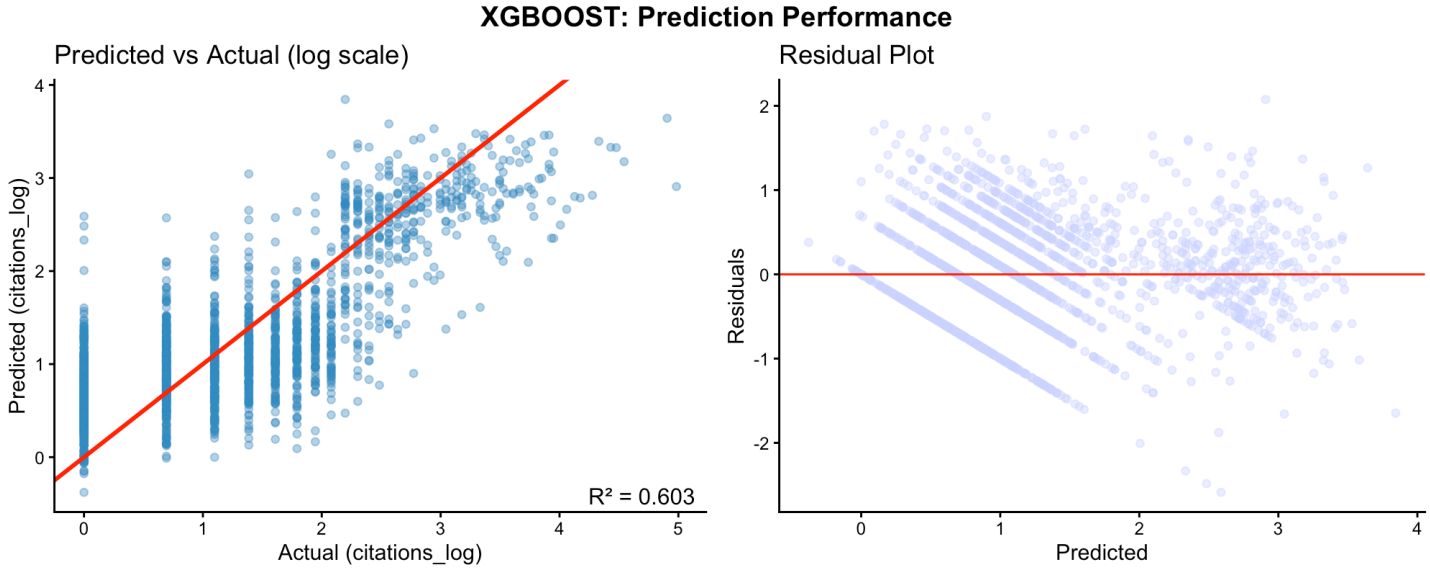
The objective is to predict the number of citations, which serves as a proxy for academic impact and helps explain why some papers attract more scholarly attention than others. Several hypotheses are proposed to explain citation counts:

- Recency: It is probable that older papers are expected to accumulate more citations simply due to longer exposure. Accordingly, this variable is computed as the difference between the publication year and the most recent year in the dataset.
- Num_authors: The total number of contributing authors is extracted from the authors field after preprocessing. It is hypothesised that papers with more authors may receive more citations due to broader academic networks.
- N_famous_authors: Some authors may have higher visibility or reputation. This is measured by computing the average number of views for each author across their publications; authors in the top 20% are classified as prominent.
- Diversity: Topic diversity is derived from LDA topic proportions using Shannon entropy (Yan et al., 2011). Higher entropy indicates broader topical coverage, which may increase relevance to a wider audience and lead to more citations.
- Is_altmetric: A binary indicator is used to capture whether a paper has non-zero Altmetric activity, reflecting public or media attention. Binarisation is applied due to the high proportion of zero values in this particular variable.
- Views: The number of views is included as a predictor and log-transformed to address heavy right skewness.
- Pages: The number of pages is included as a proxy for content depth.
- Topic: The dominant topic from the LDA model is assigned to each paper based on the highest topic probability.
- Journal: Journal information is taken directly from the dataset and encoded using one-hot encoding.

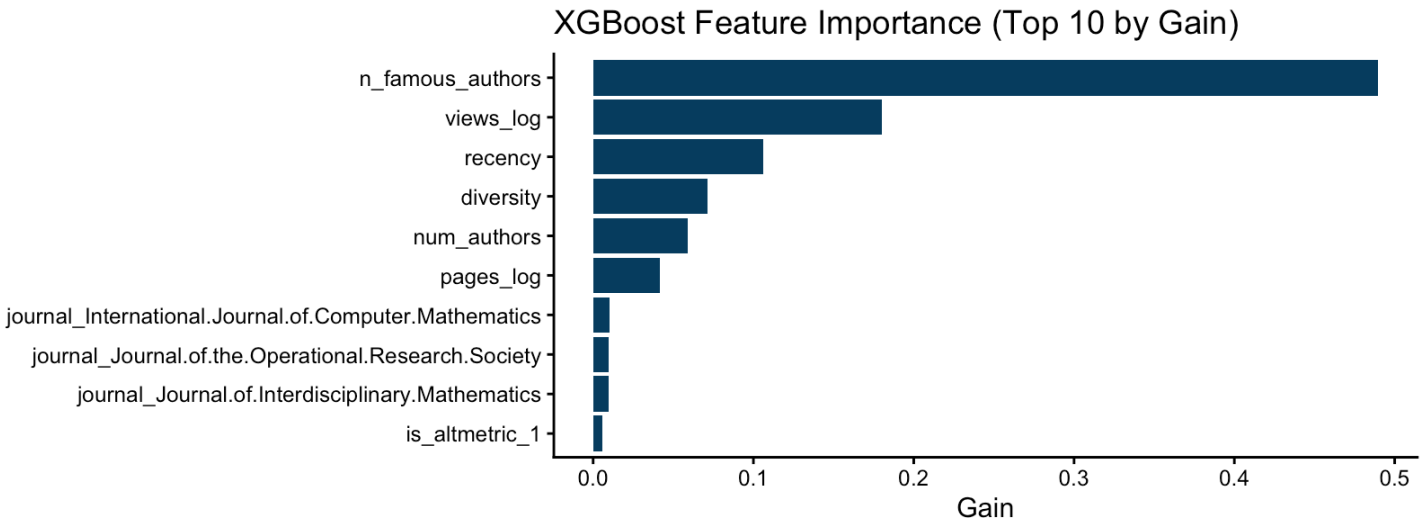
Citation counts are predicted using Linear Regression, Random Forest, and XGBoost, with citations log-transformed to reduce skewness. XGBoost performs best, explaining about 60% of the variance, while Random Forest regularisation lowers R^2 from 0.65 (overfitting) to 0.53 to mitigate overfitting and improve generalisation. Linear Regression performs worst, indicating that citation behaviour is not well captured by linear assumptions, thereby supporting H2.

Model	R_squared_test	RMSE_test	MAE_test
Multiple Linear Regression	51.3%	0.72	0.58
Random Forest	53.8%	0.70	0.57
XGBoost	60.3%	0.65	0.53

The predicted-versus-actual plot shows a clear positive association, but with substantial prediction errors. Further improvements in explanatory power may require incorporating additional factors not captured in the current models, such as journal reputation, topical timeliness, publication accessibility, and author roles (lead versus co-authors).



XGBoost shows that citation prediction is driven primarily by author reputation, visibility, and recency, which together account for about 80% of the model's gain, while journal-related features contribute negligibly. In contrast, Linear Regression places greater weight on journal variables due to its linear assumptions, suggesting that journal affiliation is less influential once non-linear effects are captured.



5.3 Classification

The objective is to predict a paper's journal membership using both textual and numerical features. Support Vector Machines (SVM) and Multinomial Logistic Regression are employed as classification models.

Textual features are extracted from abstracts and represented using TF–IDF to weight terms by their discriminative power across documents. To prevent data leakage, the TF–IDF representation is fitted on the training set only and subsequently applied to the test set.

Scaled numerical variables, including page count, views, citations, and Altmetric score, are incorporated as predictors to capture differences in paper length, readership, scholarly impact, and public visibility, all of which may vary across journals and contribute to journal-level differentiation.

SVM model receives higher performance as it effectively handles high-dimensional sparse text data and finds optimal maximum-margin decision boundaries between classes.

Model	Accuracy	Kappa	Sensitivity	Specificity	Precision	F1
glmnet	0.86	0.80	0.82	0.95	0.85	0.83
SVM	0.89	0.84	0.85	0.96	0.88	0.86

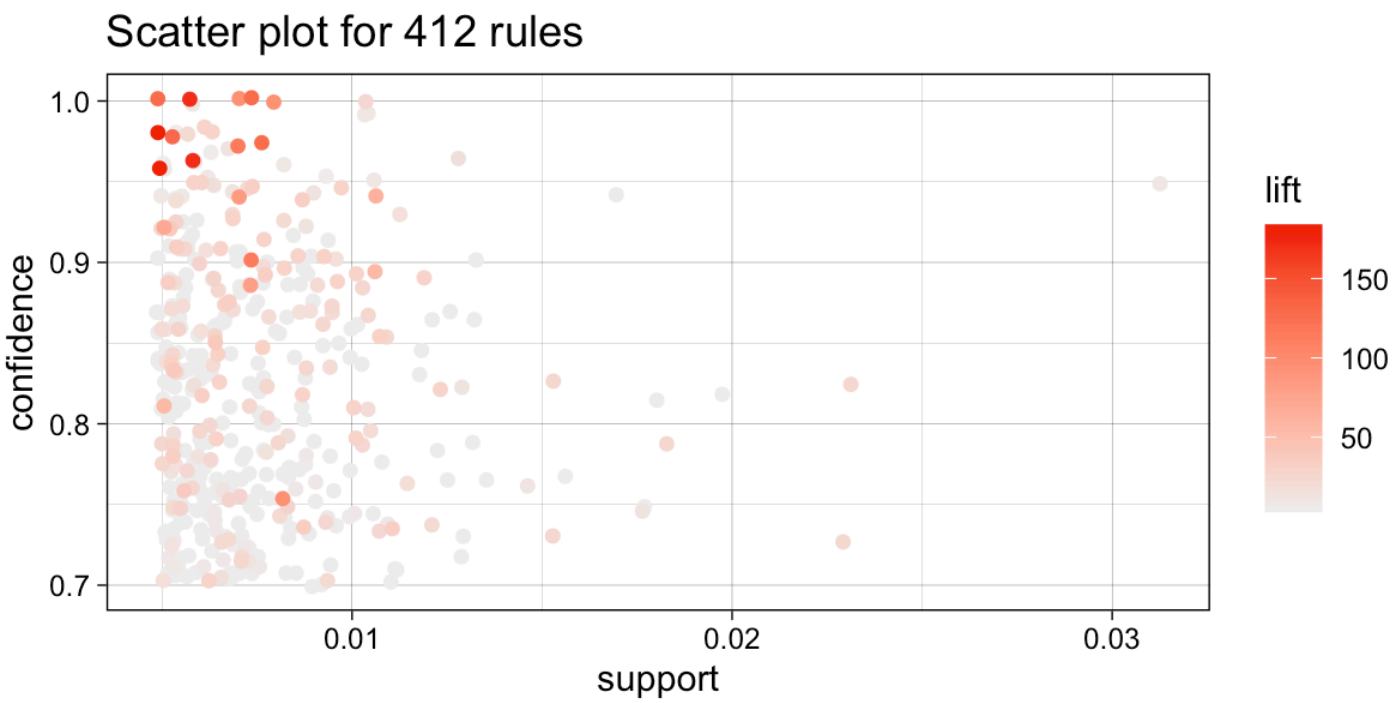
Based on the SVM confusion matrix, the Journal of the Operational Research Society outperforms others in sensitivity (96.7%), reflecting its distinctive domain-specific vocabulary (supply chain, scheduling, optimisation), while the Journal of Interdisciplinary Mathematics performs lower (72.5%). A possible explanation for this may be due to its broad, overlapping subtopics that share terminology with multiple other journals.

The SVM classifier achieves high journal-specific specificity (above 90%), while sensitivity varies across journals: papers with ambiguous content are often assigned to the Journal of the Operational Research Society, whereas the Journal of Interdisciplinary Mathematics is predicted only when strong and distinctive signals are present.

Note: JORS = Journal of the Operational Research Society; JIM = Journal of Interdisciplinary Mathematics; IJCM = International Journal of Computer Mathematics; AAI = Applied Artificial Intelligence.

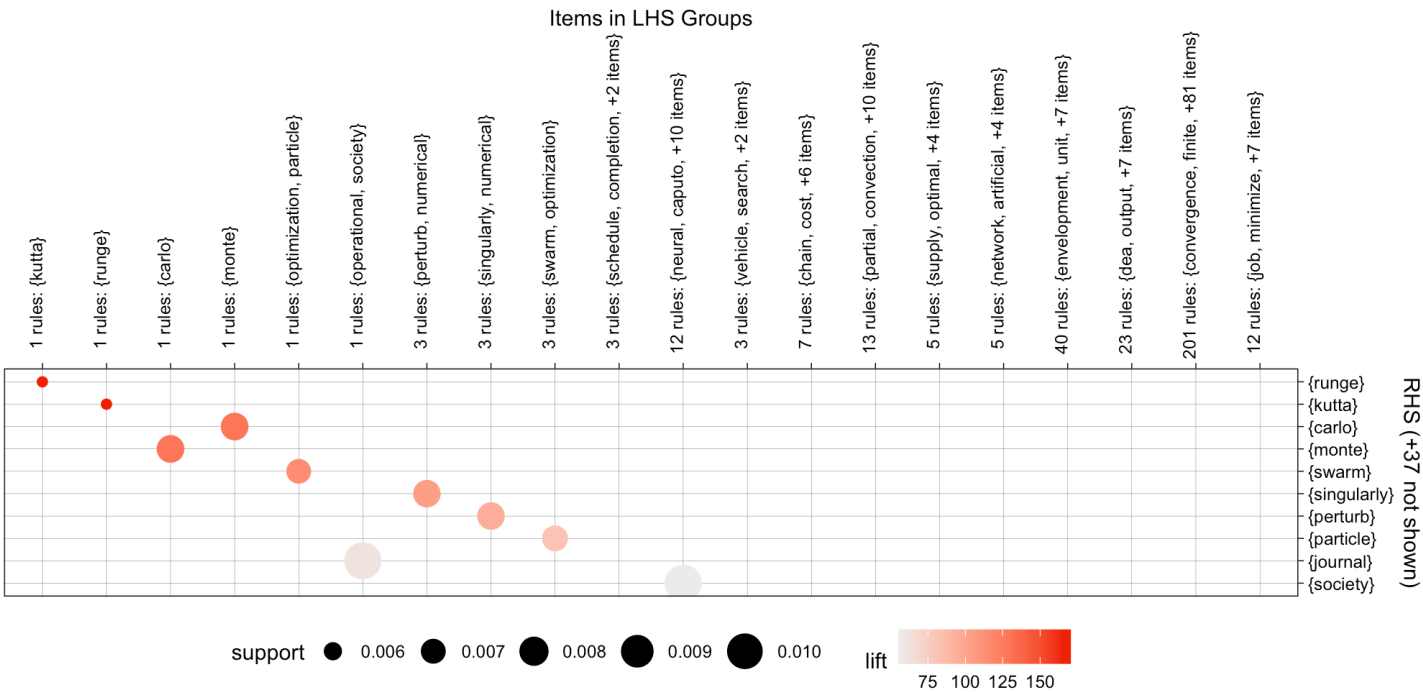
5.4 Association

The scatter plot of association rules reveals that most rules have low support but high confidence, which is typical for text-based transactions. A subset of rules exhibits very high lift values, indicating strong and non-random co-occurrence patterns. These rules are selected for further qualitative interpretation.



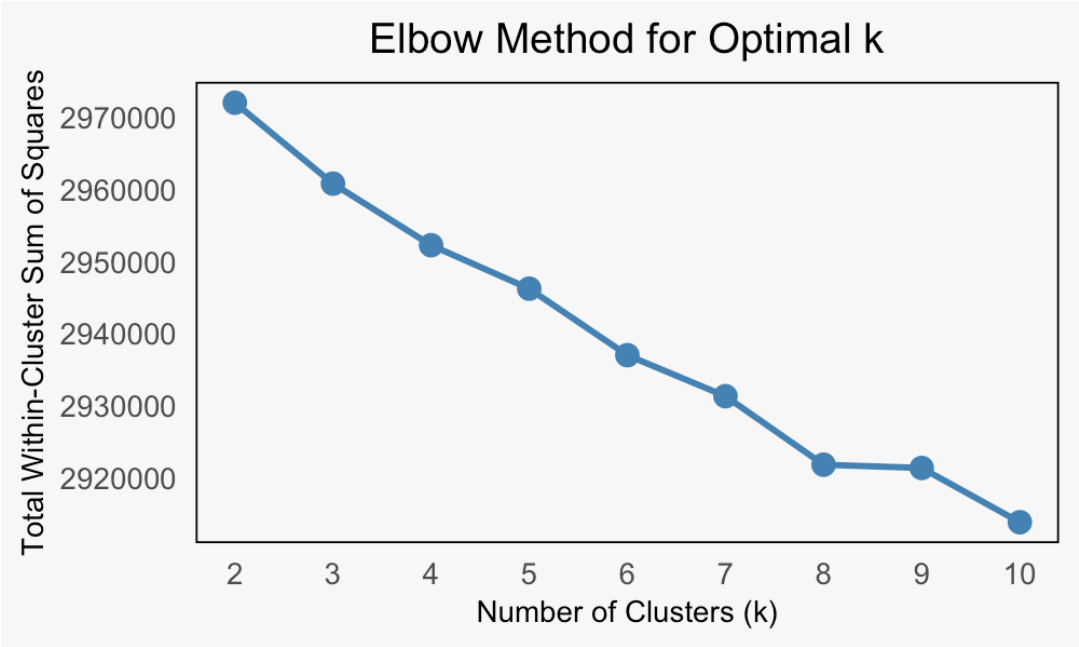
To highlight the most informative patterns, only rules with a lift greater than 10 and a frequency above 50 were retained, and the top 100 rules were analyzed. As a result, the network splits into two disconnected components. What can be clearly seen in this graph is that cluster 1 is centered on “efficiency”, “performance”, “job schedule”, and “decision-making”, focusing on practical operational optimization, whereas cluster 2 is anchored by “optimization”,”genetic”, “neural network” and”simulate”, focusing on algorithmic and mathematical tools for solving AI and computational problems.

```
Registered S3 methods overwritten by 'registry':
  method               from
  print.registry_field proxy
  print.registry_entry proxy
```

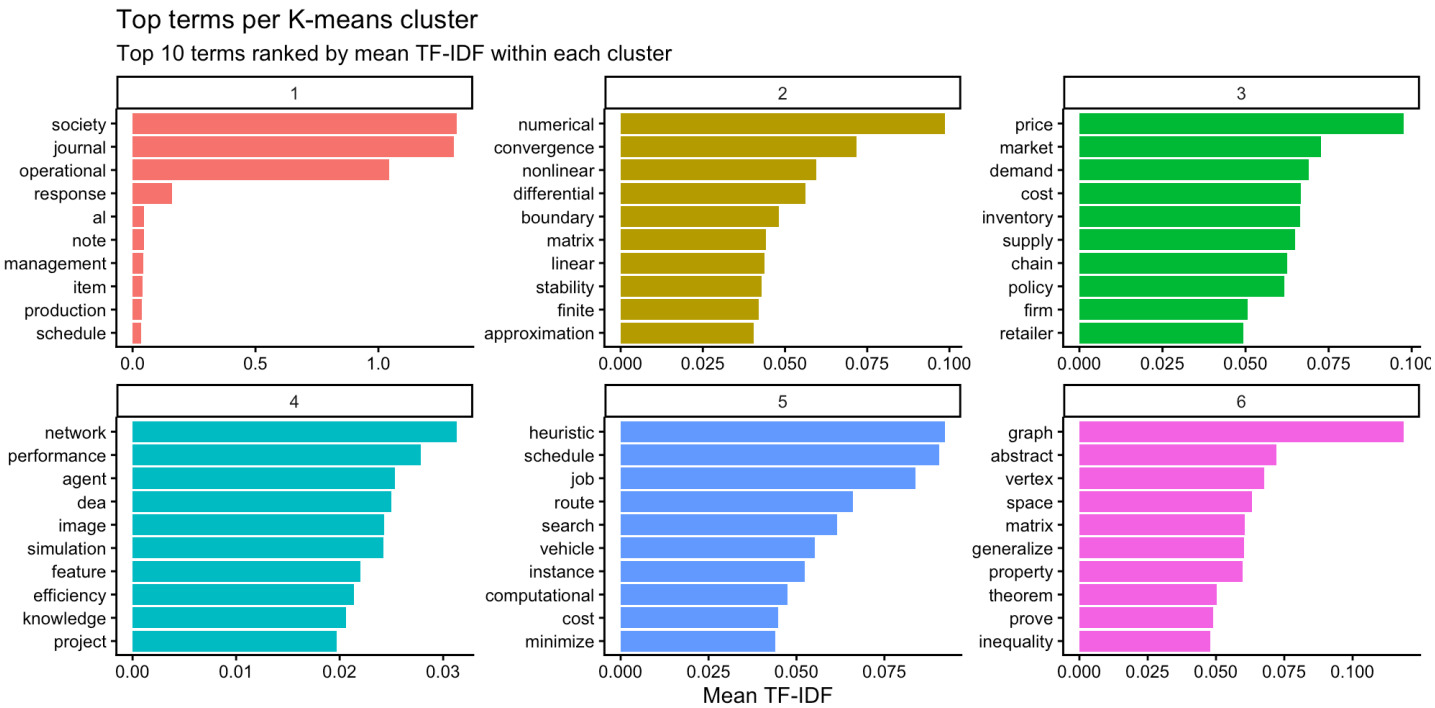


5.5 Clustering

Documents were represented using TF–IDF vectors, following common practice in previous sections. The optimal number of clusters was selected using the elbow method, and K-means clustering was applied with K = 6.

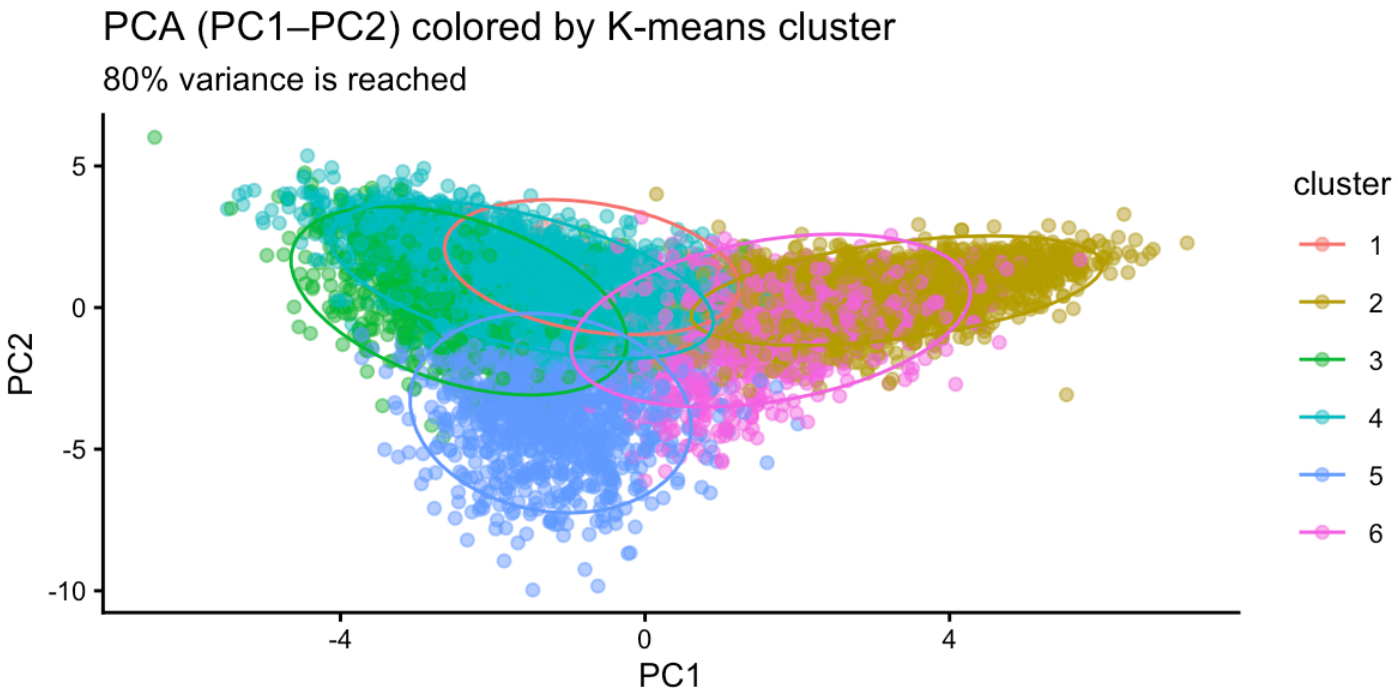


The K-means clusters reveal distinct thematic groupings driven by dominant vocabularies. Clusters 2 and 6 are characterised by numerical and mathematical terminology, while clusters 3 and 5 focus on economic, supply chain, and optimisation-related concepts. In contrast, clusters 1 and 4 capture journal-oriented and AI/agent-based research themes, indicating that TF–IDF-based clustering can effectively separate documents by methodological and application domains.



5.6 PCA for Clustering

PCA projection shows that while some overlap exists, several clusters exhibit coherent separation, suggesting meaningful latent structure in the document corpus.



6 Evaluation & discussion

6.1 Summary of findings

H1 (Thematic Evolution) A redistribution of performance evaluation topics and a pronounced increase in numerical analysis, particularly within the Journal of Interdisciplinary Mathematics, indicate a clear thematic shift after 2015

H2 (Non-linear prediction for citations) XGBoost ($R^2 = 60.3\%$) and feature importance analysis show that citation impact is driven primarily by author reputation, visibility, and recency, while journal-related features contribute negligibly. Linear Regression fails to capture these relationships, confirming the non-linear nature of citation behaviour.

Supporting analyses Optimisation-focused journals are easier to classify, whereas interdisciplinary journals exhibit lower separability due to overlapping vocabulary. Association rule mining reveals strong, domain-specific word co-occurrence with limited cross-domain overlap, and clustering identifies distinct yet partially overlapping thematic groups. Together, these results highlight blurred thematic boundaries consistent with interdisciplinary research and complex citation dynamics.

6.2 Limitations

The analysis is descriptive, not causal. Bag-of-words ignores semantic context. Journal imbalance (73% from JORS/IJCM) may affect generalisability. External factors such as open access, self-citation, and other relevant variables are not captured.

6.3 Conclusions

While research themes have shifted toward performance evaluation and numerical methods over time, citation success is driven primarily by author reputation and visibility rather than content alone.

7 References

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